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晶圓廠 晶舟變形檢驗機操作規範

FAB Wafer Deformation Inspection O.I.

壹、目的 (Purpose)

貳、適用時機及適用範圍

Applicable Timing and Scope of Application

參、內容(Content)

一、規格及程式內容

Recipe name and specification

二、設備機台 (Equipment)

三、操作步驟 (Operation Steps)

肆、例行作業 (Routine Operations)

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晶舟變形檢驗機操作規範

Wafer Deformation Inspection O.I.

壹、目的(Purpose):

製程目的：利用雷射光掃描標準片以檢驗晶舟 (PP及 TEFLON) 是否變形。

Purpose: Use laser beam to scan the standard wafer in order to check for deformation of the wafer cassette (PP and Teflon).

貳、適用時機及適用範圍(Applicable Timing and Scope of Application):

晶舟清洗完之檢驗。

Inspection after washing wafer cassette.

參、內容(Content)

一、程式名稱及規格(Recipe name and specification):

[Wafer carrier distortion analyzer]

Registration of Product Standard

Product #	: AA STD	File name	: AA.STD
Product name	: AA STD		
Display mode	:	1	
Wafer Size	: [inch]	6	
Number of pocket	:	25	
Wafer pitch	: [mm]	4.76	
First wafer height	: [mm]	13.3	Remedy : [mm] 1.0
H bar height	: [mm]	8.5	Remedy : [mm] 0.8
Remedy of top position	: [mm]	0.5	
Remedy of bottom position	: [mm]	0.5	
Remedy of slant position	: [mm]	1.33	

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[Wafer carrier distortion analyzer]

Registration of Product Standard

Product #	: AB STD	File name	: AB.STD
Product name	: AB STD		
Display mode	:	1	
Wafer Size	: [inch]	6	
Number of pocket	:	25	
Wafer pitch	: [mm]	4.763	
First wafer height	: [mm]	13.3	Remedy : [mm] 1.0
H bar height	: [mm]	8.5	Remedy : [mm] 0.8
Remedy of top position	: [mm]	0.5	
Remedy of bottom position	: [mm]	0.5	
Remedy of slant position	: [mm]	1.33	

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[Wafer carrier distortion analyzer]

Registration of Product Standard

Product #	: EQT STD	File name	: EQT.STD
Product name	: EQT STD		
Display mode	:	1	
Wafer Size	: [inch]	6	
Number of pocket	:	25	
Wafer pitch	: [mm]	4.76	
First wafer height	: [mm]	13.3	Remedy : [mm] 0.7
H bar height	: [mm]	8.5	Remedy : [mm] 0.8
Remedy of top position	: [mm]	0.5	
Remedy of bottom position	: [mm]	0.5	
Remedy of slant position	: [mm]	1.0	

[Wafer carrier distortion analyzer]

Registration of Product Standard

Product #	: EQP STD	File name	: EQP.STD
Product name	: EQP STD		
Display mode	:	1	
Wafer Size	: [inch]	6	
Number of pocket	:	25	
Wafer pitch	: [mm]	4.76	
First wafer height	: [mm]	13.3	Remedy : [mm] 0.8
H bar height	: [mm]	8.5	Remedy : [mm] 0.8
Remedy of top position	: [mm]	0.5	
Remedy of bottom position	: [mm]	0.5	
Remedy of slant position	: [mm]	1.0	

二、設備機台(Equipment)

本機台由 KYOEI SEMICONDUCTOR 公司製造，型號 CA-008。

This machine is manufactured by KYOEI SEMICONDUCTOR, model no. CA-008.

三、操作步驟(Operation Steps):

1 開機(Power on)

1.1 打開 CA008 機台電源（開關在機台正面左下方）

Turn on the power to the CA008 machine (switch is located on the lower left side on the front of the machine)

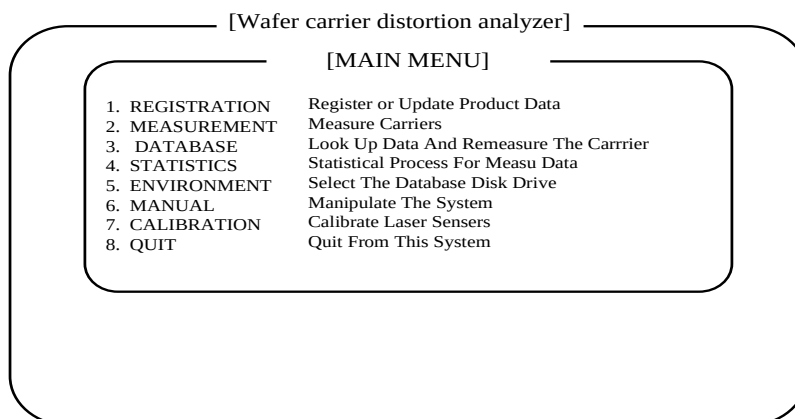
1.2 打開電腦及螢幕之電源。

Turn on the power to the computer and monitor.

1.3 等待約三分鐘，螢幕上出現如下之畫面。

Wait for 3 minutes and the screen displays this.

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2.以上下方向鍵將反白區移至 2.MEASUREMENT 後，再按 ENTER 即進入以下螢幕。

[Wafer carrier distortion analyzer]

NO.	File Name	Product #	Product Name	Total	5
1	AA.std	EA	EA	3/01/1996	
2	AB.std	EA	EA	3/01/1996	
3	EQT.std	EA	EA	3/01/1996	
4	EQP.std	EA	EA	3/01/1996	

Quit Select Print
F1 F2 F3

Use the up and down arrows to move the highlighted (white) area to 2. MEASUREMENT and then press ENTER to enter the following screen.

3.量測 TEFLON 晶舟時以上下方向鍵時反白區請至 AA.std 後按 Enter 進入以下畫面。

量測 PP 晶舟時，以上下方向鍵將反白區移至 AB.std 後按 Enter 進入以下畫面。

量測爐管區專用 TEFLON 晶舟時以上下方向鍵時反白區請至 EQT.std後按 Enter 進入以下畫面。

量測專用 PP 晶舟時以上下方向鍵時反白區請至 EQP.std後按 Enter 進入以下畫面。

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量測時 晶片置於晶舟時(Notch)需朝下。

To measure the TEFLON cassette, use the up and down arrow keys to move the highlighted (white) area to AA.std and then press ENTER to enter the following screen.

To measure the PP cassette, use the up and down arrow keys to move the highlighted (white) area to AB.std and then press ENTER to enter the following screen.

[Wafer carrier distortion analyzer]

NO.	Lot#	Record #	Qty.	Date	Auther	Comment
New registration						

Quit Select Print
F1 F2 F3

4.以上下方向鍵將反白區移至 1, 再按 ENTER 進入以下畫面。

Use the up and down arrow keys to move the highlighted (white) area to 1 and then press ENTER to enter the following screen.

[Wafer carrier distortion analyzer]

File name: Record#: 000001 Author: 8/19/1993 15:53
 Product#: Remark#: Lot#:
 Product name: Comment:

Quit Select Print
F1 F2 F3

5.以左右方向鍵將反白區移至以下各位置並 key 入以下各項：

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Use the left and right arrow keys to move the highlighted (white) area to the following locations and key in each item:

項目 (Item)	需要輸入之內容_(Contents that needs to key in)
Filname	1
Lot#	1
Auther	1
Remark	1
Comment	1

6.將 25 片標準片以晶舟轉換器送入欲量測之晶舟內，H-Bar 向下放至 CA-008 機台

Stage 上推平整臂至標準片上，以確保標準片在定位後移出平整臂。

Use the cassette converter to insert the 25 pieces of standard wafers into the cassette to be measured. The H-Bar should be placed faced down on the CA-008 machine. Push the arm over the standard wafers on the Stage to assure that the arm will move out after standard wafers are properly fixed.

7.在 CA-008 機台上按下 Start，即開始量測，再由螢幕上得知該 Cassette 是 GOOD 還是 POOR，再將此 Cassette 放至 GOOD 區或 POOR 區定位。不合格之晶舟覆驗 2 次確認不合格就報廢。
Press Start under the CA-008 machine to start measuring and the screen will display whether the Cassette is GOOD or POOR and place it in the GOOD or POOR area accordingly.

8.如須連續檢驗相同顏色的晶舟時選 ReTest 後按 ENTER，在 CA-008 機台上按下 Start，即開始量測。

肆、例行作業(Routine Operations)

一、每天日班負責洗晶舟盒之 EA，於上班初以無纖維布擦拭機台。

The day shift EA in charge of the daily washing of the cassette box should use a fiberless cloth to wipe the machine at the start of the shift.

二、晶舟清洗前需檢驗，晶舟以晶舟變形檢驗機量測晶舟是否變形如 GOOD 送至BOX-CLEAN 機台清洗。並填寫BOX-CLEAN 機台 RUN 貨及晶舟檢驗紀錄表。(如附件)

三、標準 WAFER 及 PP，TEFLON 標準晶舟放置於機台旁之盒內，鑰匙由 A-TUNNEL 各班領班負責管理，標準 WAFER 請 EA 於每個月的第一個及第三個星期一，日班上班時以 RCA 機台之 WASH DOWN 程式清洗。

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Place the Standard wafer, PP and TEFLON standard cassette in the box beside the machine. Each shift leader in A-TUNNEL is in charge of keeping the keys. The EA should use the WASH DOWN recipe to wash the standard WAFER in the RCA machine during the day shift in the first and third Mondays of each month.

四、當PP 或 TEFLON 晶舟檢驗記錄達 140 個，即洽 PE 清除部分存檔刪除，選定 *.db

檔刪除，注意不可刪除 *.std，*.std 檔為檢驗用之標準。方式請參閱附件二。

When the number of PP or TEFLON cassette inspection record reaches 140, consult the PE to clear up the file, Select and delete the *.db files. Make sure that you must not delete the *.std files. The *.std files are for inspection. Refer to Attachment 2 on how to delete file records.

五、產品下線前晶舟需全檢，合格之晶舟再以洗盒機清洗，清洗完之晶舟再提供給 S130 下線使用。

Cassette have to be check by the cassette deformation inspection machine before wafer start, then use the box cleaner to clean it.

When the cassette clean completely then offer S130 to do the wafer start using..

六、每週週一做 PP，TEFLON 標準晶舟之檢驗，連續量測 3 次如有一次不合規格洽設備工程師調整機台。

伍、品質控制(Quality Control)

一、機台每半年須由設備工程師做一次校正及檢驗。

The equipment engineer must calibrate and inspect this machine once every 6 months.

檢驗項目有(Inspection items include):

1.以標準片用“STANDARD 程式對 Laser Stage 與 C Stage 校正，其 Standard Height 為 25300 ~ 25600(μm) 之間。

Use the standard wafer to calibrate the Laser Stage and C Stage using "Standard" recipe. The standard height should be between 25300 ~ 25600 (μm).

2.以兩支 Pin Gauge 放於 C Stage 上量測 Laser Sensor 的敏感度(Sensitivity) Spec 為 2000±5μm。

Place 2 Pin Gauge on C Stage to measure the sensitivity of the Laser Sensor. Spec is 2000±5μm.

陸、生效與修訂

本規範之公佈實施及其修訂核准層級皆依會簽/核決/分發依循範例為之。

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柒、附件(Attachments)

附件一：BOX-CLEAN 機台 RUN 貨及晶舟檢驗紀錄表。

附件二：CDA 晶舟變形檢驗機台 HARDDISK 清除不要檔案，操作程序。

Attachment 2: Procedure for deleting unneeded files in the CDA Cassette Deformation Inspection
Machine Hard Disk

附件一(Attachment 1)

[illegible]

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附件一~一(Attachment 1.1)

[illegible]

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附件一~二(Attachment 1.2)

表格填寫說明

(Explanation for filling the form)

表格名稱：BOX-CLEAN 機台 RUN 貨及晶舟檢驗紀錄表

項次	欄 位 名 稱	填 寫 方 式 及 內 容 描
Item	Field Name	Filling Method and Description of Content

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1	日期 Date	填寫量測當時日期，如：月 / 日 Date of measurement such as month/day
2	班別 Shift	填寫班別 Write Shift
3	工號 Work no.	記錄 EA 工號 Record EA work no.
4	時間 Time	晶舟輸入及輸出機台的時間。 Record the time of wafer load and unload
5	盒子種類(紅、藍、黃...)	填寫(紅色、藍色、黃色...處打 V
6	晶舟種類(PP或Telon或 SPUTTER 專用晶舟)	填寫是藍色晶舟(PP)或白色晶舟(Teflon)或 (SPUTTER 專用晶舟處打 V)
7	晶舟檢驗結果 晶舟編號 Cassette no. GOOD GOOD POOR POOR 備註欄(異常記錄) Remarks (Abnormal Record)	填寫晶舟檢驗結果 填寫該晶舟號碼 Write down cassette no. 結果是 GOOD，請打勾 Check if result is GOOD 結果是 POOR，請打勾 Check if result is POOR 有異常狀況時註明 Note down abnormal condition

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附件二 CDA 晶舟變形檢驗機台 HARDDISK 清除不要檔案，操作程序!

Attachment 2: Procedure for deleting unneeded files in the CDA Cassette Deformation Inspection Machine Hard Disk

1 避免誤殺必要檔案，使PE再而三的重建。

Avoid deleting files accidentally to save the PE from having to rebuild the files repeatedly.

2 如果機台顯示空間不足無法 SAVE 量測結果，依下列步驟進行。

If the system shows that there is not enough space to save the measurement results, follow these steps:

3 在量測畫面 按 QUIT 到 DOS 畫面。

Under the Measurement Menu, click on QUIT to return to the DOS Menu.

4 在 A:TESTER> 輸入 CD..

At A:TESTER>, key in CD..

5 在 A:> 輸入 WIN 按 ENTER.

At A:>, key in WIN and press ENTER.

6 打開檔案管理員，打開DATABASE檔案夾。

Open the Programs and then Open the DATABASE folder.

7 選定 *.db 檔 刪除，注意不可刪除，*.std, *.std 檔為檢驗用之標準。

Select and delete the *.db files. Make sure that you don't delete the *.std files. The *.std files contain the standards for inspection.

8 完畢後即離開window畫面。

When finished, exit the Windows screen.

9 A:> 輸入 CD TESTER.

At A:>, key in CD TESTER.

10 A:TESTER> 輸入 TESTER 回操作畫面。

At A:TESTER>, key in TESTER to return to the Operation Menu.